

- Class
- Functions

GOC - Game Object Composition
 GOCId - unsigned int
 ComponentTypeId - enum

[ComponentTypeIds.h]

GameObjectFactory [Factory.h & Factory.cpp]

- GameObjectFactory
 - ~GameObjectFactory
 - GOC* Create (const std::string& filename)
 - void Destroy (GOC* gameObject)
 - virtual void Update (float dt)
 - virtual std::string GetName()
 - virtual void SendMessage (Message* message)
 - void DestroyAllObjects()
 - GOC* CreateEmptyComposition()
 - GOC* BuildAndSerialize (const std::string& filename)
 - void IdGameObject (GOC* gameObject)
 - void AddComponentCreator (const std::string& name, ComponentCreator* creator)
 - GOC* GetObjectWithId (GOCId id)
- Datas:
- unsigned LastGameObjectId
 - std::map<std::string, ComponentCreator*> GameObjectIdMap
 - std::set<GOC*> ObjectsToBeDeleted
- extern GameObjectFactory* FACTORY

1. Create empty game obj
2. Build & Serialize game obj
3. Store game obj in map
4. Find game obj using ID
5. Delete game obj
6. Binding component with component name

GameComponent [Component.h]

- friend class GameObjectComposition
 - virtual void Initialize()
 - virtual void SendMessage (Message*)
 - virtual void Serialize (ISerializer& str)
 - GOC* GetOwner()
 - ComponentTypeId TypeId
- Datas:
- GOC* Base

ComponentCreator [ComponentCreator.h]

- ComponentCreator (ComponentTypeId typeId): TypeId (typeId)
- ComponentTypeId TypeId
- virtual ~ComponentCreator()

Functions to optimize
 GameObjectFactory.

ComponentCreatorType: public ComponentCreator [ComponentCreator.h]

- ComponentCreatorType (ComponentTypeId typeId): ComponentCreator (typeId)
- virtual GameComponent* Create()

GameObjectComposition [Composition.h & .cpp]

- friend class GameObjectFactory
 - void SendMessage (Message* message)
 - GameComponent* GetComponent (ComponentTypeId typeId)
 - type* GetComponentType (ComponentTypeId typeId)
 - void Initialize()
 - void Destroy()
 - void AddComponent (ComponentTypeId typeId, GameComponent* component);
 - GOCId GetId() & return ObjectId
- Datas:
- Component* array Components
 - GOCId ObjectId
 - GameObjectComposition
 - ~GameObjectComposition

Functions to optimize
 GameObjectFactory.

** Brief

- Pass in data thru serialization
- Use data to initialize game object type & data